

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Let $f(x) = x^3 - 3x + 1$. First find all critical numbers of f , and then find the absolute maximum and absolute minimum values of f on the interval $[0, 3]$.

Solution: $f'(x) = 3x^2 - 3$

$$f'(x) = 0 \Rightarrow 3x^2 - 3 = 0$$

$$3(x^2 - 1) = 0$$

$$3(x-1)(x+1) = 0$$

$$x = 1, -1 \rightarrow \text{two critical numbers: } 1, -1$$

$$f(1) = 1^3 - 3 \cdot 1 + 1 = -1, \quad f(-1) = (-1)^3 - 3 \cdot (-1) + 1 = 3$$

Two end points $f(0) = 1, \quad f(3) = 19$

Thus, absolute maximum value is 19.

absolute minimum value is -1.

↘ not considered,
since outside
of $[0, 3]$

Problem 2. (4 points) If $f(1) = 5$ and $f'(x) \leq 3$ for all x , how large can $f(5)$ possibly be? (using the Mean Value Theorem.)

Solution: According to Mean Value Theorem, we have.

$$f(5) - f(1) = f'(c)(5-1) \quad \text{for some } c \text{ in } (1, 5)$$

$$f(5) = f(1) + f'(c)(5-1)$$

$$= 5 + f'(c) \cdot 4$$

$$\leq 5 + 3 \cdot 4$$

$$= 17$$